

For SAR Lab 7, the last lab, we will be looking at a case study of a landslide in Sishuan Province, China.

Here is a paper describing a PS-based time series analysis for the time preceding the landslide: <https://link.springer.com/article/10.1007/s10346-017-0915-7>

You will be recreating this study using SBAS on publicly available pre-processed interferograms.

The COMET-LiCS project (Centre for the Observation and Modelling of Earthquakes, Volcanoes, and Tectonics; Looking into the Continents from Space) has processed and unwrapped 2.500.000 interferograms covering tectonically and volcanically active regions.

You may need to create a new conda environment for licsar. I did this in a couple of steps:

1. `conda create --name licsar -c conda-forge python=3.11 pip ipykernel jupyterlab numpy scipy matplotlib pandas requests beautifulsoup4 tqdm h5py`
2. `conda activate licsar`
3. `conda install -c conda-forge rasterio rioxarray xarray netcdf4 pyproj shapely`
4. `conda install -c conda-forge geopandas fiona cartopy`

The script `download_licsar_pre_event.py` downloads all interferograms prior to the event within a certain temporal baseline. To run, change the `--out` to your own directory, e.g.:

```
python download_licsar_pre_event.py --base-url
https://gws-access.jasmin.ac.uk/public/nceo_geohazards/LiCSAR_products/62/062D_05831_13
1313/ --out YOUR_LICSAR_DIRECTORY --cutoff 20170624 --max-days 72 --products unw cc
png
```

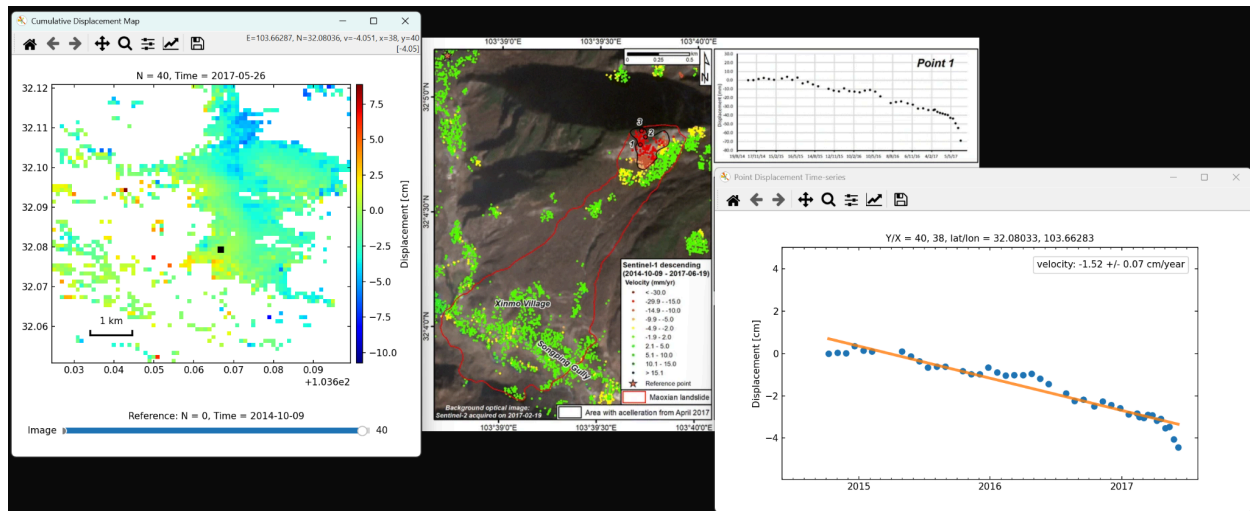
The script `licsar_to_mintpy_h5.py` converts the individual interferograms into a stack that mintpy can ingest, and subsets to a bbox AOI. To run, e.g.:

```
python licsar_to_mintpy_h5.py --licsar-dir YOUR_LICSAR_DIRECTORY --out
YOUR_MINTPY_DIRECTORY --bbox left_lon bottom_lat right_lon top_lat
```

Then you can run mintpy:

```
smallbaselineApp.py smallbaselineApp_licsar.cfg --start modify_network
```

At first, use an “auto” reference point. However, you may notice a strong seasonal signal near the landslide (use `tsview.py` from minted to find the landslide pixels). You can mitigate this by selecting a reference point close to and just outside of the landslide (change the reference Y/X point to one of your choosing in the `.cfg` file). The figure here shows my result (The pixel I selected here is the blue pixel to the left of the black reference point square in the left figure):



Note here that the pre-computed interferograms do not have every possible SAR acquisition in the database - they are typically only done every few weeks. Here, the last few points in the paper's figure were not processed by COMET-LiCSAR.

Lastly for the SAR part, find and examine the coherence file for the interferogram covering the landslide (you can find it in the LiCS database as the .cc file):

https://gws-access.jasmin.ac.uk/public/nceo_geohazards/LiCSAR_products/62/062D_05831_131313/). Do you think coherence is a useful way to map the landslide in this case? Then, in google earth engine, examine the sentinel-1 GRD files for the dates before and after the landslide. Compute how the vv has changed from before to after. Do you think this is a better, worse, or the same at mapping the landslide vs coherence loss?

Next, we will be examining the landslide with optical data. Find a pre-and post- event sentinel-2 image that has cloud-free coverage over the area affected by the landslide. You have two goals from the optical side. The first will be to map out the affected area. The second is to use a hyperspectral index or indices of your choice (NVDI, BSI, etc.) to quantify the changes that occurred due to this landslide. Can you say something about vegetation loss, for example?

To recap:

SAR:

1. Download pre-processed COMET-LiCS interferograms.
2. Subset around landslide and convert to mintpy's stack format.
3. Choose an appropriate reference point to best analyze the slow landslide time series.
4. Try to map the landslide with coherence loss and vv change.

Optical:

1. Find appropriate cloud-free images before and after the landslide in Sentinel-2 and map the landslide extent.
2. Use a hyperspectral index or indices of your choice to try to quantify the land surface changes that occurred as a result of this landslide.

Take appropriate figures, put it into a .pdf file with a description of what steps you took and a discussion of the results, and upload that file for grading.